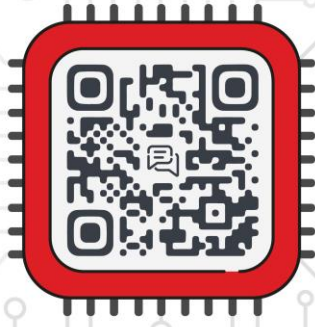




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نظري

كلية الهندسة المعلوماتية

السنة الثانية

Fourier Transforms

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RB Informatics; 1/6/2026

الاتصالات الرقمية

<3 السلام عليكم ورحمة الله وبركاته <3

Newton discovered that white light (which appears simple) is actually compound light, when it passes through the prism, it decomposes into the seven basic colors of the spectrum, the scientist "Fourier" benefited from this physical idea and applied it mathematically to signals. He considered the mathematical prism to be "Fourier analysis" which deconstructs a complex signal into its basic components.

Fourier transforms

A **Fourier Transform** is a mathematical technique that **breaks down** a complex signal into its foundational frequency components. It **shifts** data from the time or space domain into the frequency domain, allowing you to easily identify hidden frequencies, amplitudes, and phases.

For a continuous, integrable function $f(x)$, the **Fourier transform** $F(\omega)$ is formally defined as:

$$F(\omega) = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx$$

$f(x)$: The original signal in the time or space domain.

$F(\omega)$: The transformed signal in the frequency domain, returning the amplitude and phase of each frequency.

$e^{-i\omega x}$: The rotating complex exponential (Euler's formula relates this to sines and cosines).

Homework:

Explain the similarity between experiencing Newton's prism visually and mathematically using Fourier series in analyzing complex signals.

Applications

The Fourier Transform is a cornerstone of modern technology, heavily utilized across engineering, physics, and computer science:

Signal & Audio Processing: Powers noise-canceling headphones by detecting and canceling out unwanted background frequencies.

Telecommunications:

Modulates and demodulates signals in radios, Wi-Fi, and 5G/6G networks.

Medical Imaging:

Reconstructs clear images from raw data in **MRI** and **CT** scans.

MRI: Magnetic resonance imaging
CT: Computed tomography

Data Compression:

Underpins JPEG image compression and MP3 audio encoding by removing imperceptible high frequencies.

Discrete Fourier Transform (DFT):

Used for discrete, sampled signals.

Fast Fourier Transform (FFT):

A highly optimized algorithm used to compute the DFT incredibly fast. It is a standard tool found in software libraries like NumPy in Python.

Fourier series:

Used specifically for periodic signals (repeating waves) over a finite interval.

FFT

Fast Fourier transform (FFT):

When computers process digital data, they don't use the standard mathematical Fourier Transform (which is designed for continuous, infinite signals). Instead, they use the **Fast Fourier Transform (FFT)**. The FFT is a highly optimized, clever algorithm that drastically speeds up the calculation, making real-time digital signal processing possible.

Fourier transform:

This equation transfers the signal from the time domain to the frequency domain.

$$X(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$$

$x(t)$: The original signal as it appears in reality (such as a human voice or an electrical pulse).

$X(\omega)$: It is called the frequency response, and it represents the “mix” of frequencies that make up the signal.

$e^{-j\omega t}$: It is a mathematical part based on "Euler's Formula" to simplify the analysis of the signal into sinusoidal components (sine and cos)

Inverse Fourier transform:

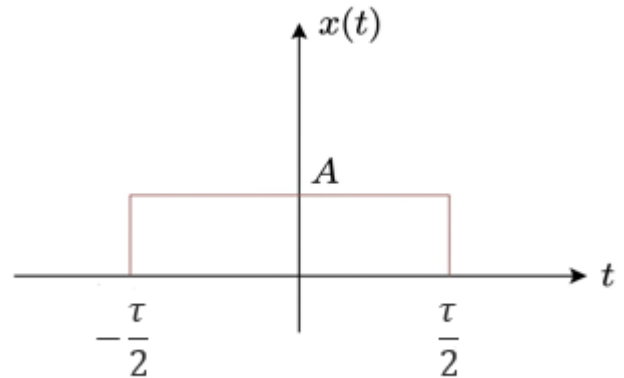
It does the exact opposite process, returning the signal from the frequency domain to the time domain

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

The goal of these equations is not to complicate the mathematics, but rather to understand the “physical idea” of how a signal is viewed from two different perspectives (time and frequency) to facilitate its processing in systems such as 5G and data compression.

Rectangular pulse (signal)

$$\begin{aligned}
 x(\omega) &= \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} A \cdot e^{-j\omega t} dt \\
 &= A \cdot \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} e^{-j\omega t} dt \\
 &= A \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} = \frac{2}{j} * \frac{A}{\omega} \left(e^{-j\frac{\omega\tau}{2}} - e^{j\frac{\omega\tau}{2}} \right) =
 \end{aligned}$$



$$\frac{2A}{\omega} * \left(\frac{e^{j\frac{\omega\tau}{2}} - e^{-j\frac{\omega\tau}{2}}}{2j} \right) = \frac{2A}{\omega} \sin \frac{\omega\tau}{2} * \frac{\omega\tau}{2} = \frac{2A}{\omega} * \frac{\tau\omega}{2} * \frac{\sin \left(\frac{\omega\tau}{2} \right)}{\frac{\omega\tau}{2}} = A\tau \cdot \frac{\sin \left(\frac{\omega\tau}{2} \right)}{\frac{\omega\tau}{2}} \rightarrow \text{Sync function}$$

يبدأ الاشتقاق بالقانون العام لتحويل فورييه $x(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$

النبضة المستطيلة معرفة فقط في المجال $\left[-\frac{\tau}{2} \rightarrow \frac{\tau}{2} \right]$ وقيمتها خارج هذا المجال هي صفر

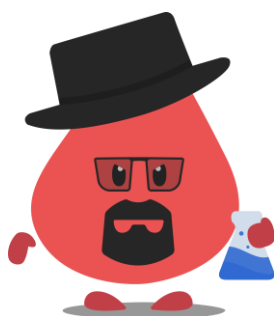
لذلك يتم تغيير حدود التكامل لتصبح $\left[-\frac{\tau}{2} \rightarrow \frac{\tau}{2} \right]$ بدلاً من اللانهاية وتستبدل $x(t)$ بقيمتها الثابتة A (amplitude).

يتم اخراج القيمة الثابتة A خارج التكامل ثم مكاملة الدالة الاسية

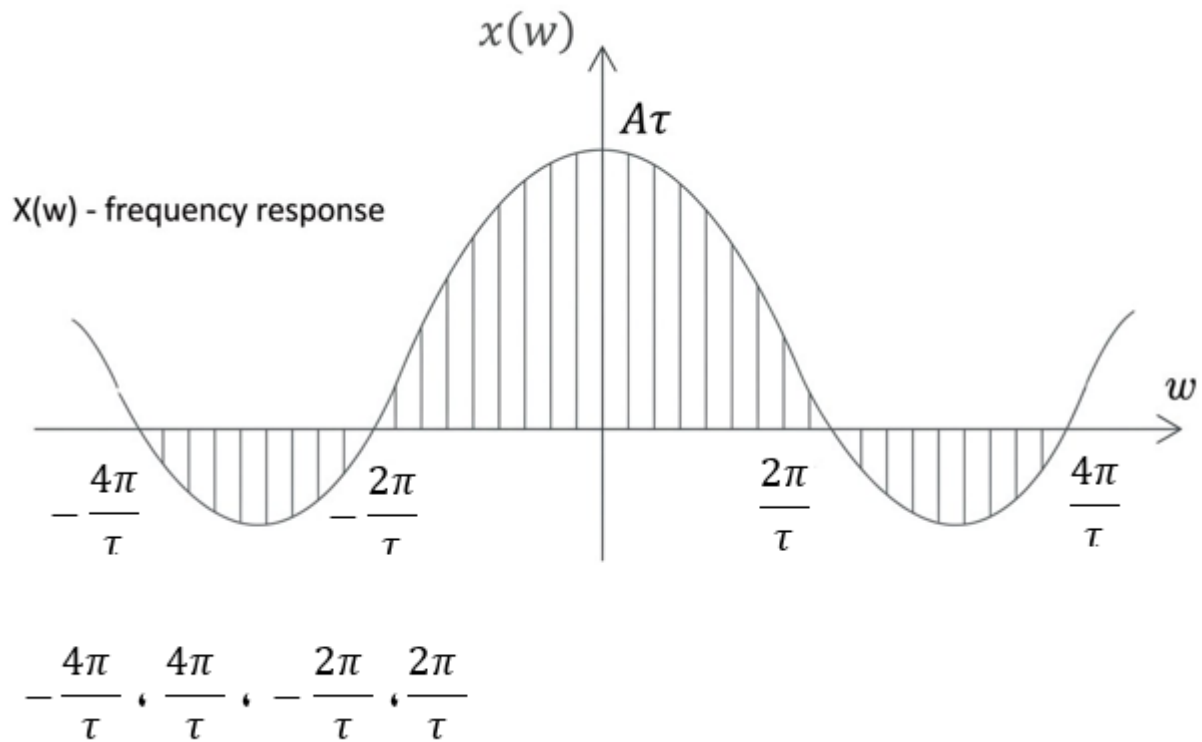
نستخدم صيغة اويلر لتحويل الاسس المركبة الى دالة ساين (sin) لتبسيط الناتج

نصل الى الشكل القياسي المعروف في الرياضيات والاتصالات $\frac{\sin \left(\frac{\omega\tau}{2} \right)}{\frac{\omega\tau}{2}}$

النتيجة النهائية هي $A\tau \cdot \frac{\sin \left(\frac{\omega\tau}{2} \right)}{\frac{\omega\tau}{2}}$ وتسمى هذه الدالة بـ *sync function*



$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$



The horizontal axis (ω): represents the angular frequency. The further we are from the center, the higher the frequency

The vertical axis ($x(\omega)$): represents the amplitude (power) of each frequency.

The maximum signal strength $A(\tau)$: is concentrated at frequency zero.

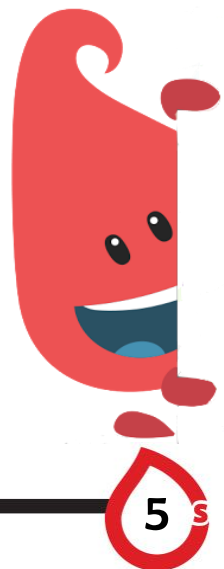
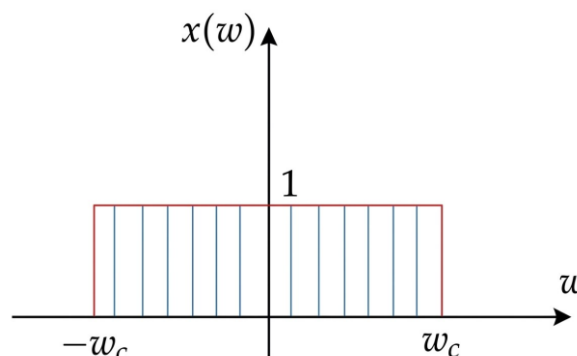
Sync function: The standard result of converting any rectangular pulse of time into frequency

Zero points: $(\pm \frac{4\pi}{\tau})$ $(\pm \frac{2\pi}{\tau})$

IFT

Inverse Fourier transform (IFT):

Return from frequency domain to time domain



Example:

$$x(\omega) = \begin{cases} 1 & -\omega_c \leq \omega_0 \leq \omega_c \\ 0 & \text{else} \end{cases}$$

$$x(t) = ?$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} x(\omega) \cdot e^{J\omega t} d\omega \rightarrow x(t) = \frac{1}{2\pi} \int_{-\omega_c}^{+\omega_c} 1 \cdot e^{J\omega t} d\omega$$

$$= \frac{1}{2\pi} * \left[\frac{e^{J\omega t}}{Jt} \right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi} * \left(\frac{e^{J\omega_c t}}{Jt} - \frac{e^{-J\omega_c t}}{Jt} \right)$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= \frac{1}{\pi t} \left(\frac{e^{J\omega_c t} - e^{-J\omega_c t}}{2J} \right)$$

$$\frac{1}{\pi t} \sin(\omega_c \cdot t) * \frac{\omega_c}{\omega_c} = \frac{\omega_c}{\pi} * \frac{\sin(\omega_c \cdot t)}{\omega_c \cdot t} \rightarrow \text{Sync function}$$

يبدأ الاشتقاق بالقانون العام لتحويل فورييه العكسي $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} x(\omega) \cdot e^{J\omega t} d\omega$ نبضة مستطيلة قيمتها ثابتة وتساوي 1 في المجال المحصور بين $[-\omega_c, \omega_c]$ ، وقيمتها خارج هذا المجال هي صفر

لذلك يتم تغيير حدود التكامل لتصبح محصورة بين $[-\omega_c, \omega_c]$ ، بدلاً من اللانهاية، ونستبدل الـ

$$x(\omega) \text{ بقيمتها الثابتة (1): } x(t) = \frac{1}{2\pi} \int_{-\omega_c}^{+\omega_c} 1 \cdot e^{J\omega t} d\omega$$

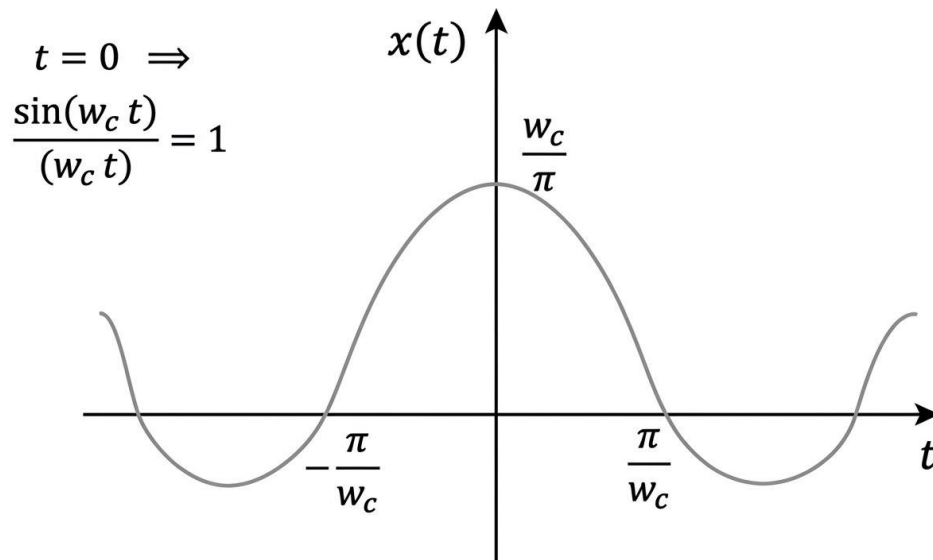
يتم مكاملة الدالة الأسية $e^{J\omega t}$ بالنسبة للمتغير ω

نستخدم صيغة اويلر لتحويل الأسس المركبة إلى دالة ساين (sin) لتبسيط الناتج

$$\text{نصل الى الشكل القياسي: } \frac{1}{\pi t} \sin(\omega_c \cdot t) * \frac{\omega_c}{\omega_c}$$

النتيجة النهائية هي $\frac{\omega_c}{\pi} * \frac{\sin(\omega_c \cdot t)}{\omega_c \cdot t}$ وتسمى هذه الدالة بـ *sync function*





Horizontal axis (t): represents the time axis

Vertical axis (x(t)): represents the signal amplitude

Maximum Peak: It is centered at the moment $t = 0$, and is exactly equal to $\frac{\omega_c}{\pi}$.

Sync function: The standard result resulting from converting an ideal rectangular pulse from the frequency domain to the time domain.

Zero crossings: the points where the signal intersects the horizontal axis ($x(t) = 0$), and is repeated at the values A

IDFT

The Inverse Discrete Fourier Transform (IDFT) is a mathematical process used in digital signal processing to convert a signal from the frequency domain back into the time domain. It is the exact reverse of the Discrete Fourier Transform (DFT), allowing you to reconstruct an original waveform after it has been analyzed.

$$x(\theta) \rightarrow x[n]$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} x(\theta) \cdot e^{jn\theta} d\theta$$

$x(\theta) = 1$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} 1 \cdot e^{jn\theta} d\theta = \dots \dots \dots = \delta[n]$$

$$\theta = \omega T = 2\pi f \cdot \frac{1}{f} = 2\pi$$

$$\rightarrow -\pi \leq \theta \leq \pi$$

يبدأ الاشتقاق بالقانون العام لتحويل فورييه العكسي للإشارات المتقطعة (IDTFT) حيث يمثل $x[n]$ الإشارة في المجال الزمني المتقطع، و $x[\theta]$ هو الطيف الترددي المستمر والدوري. نبضة مستطيلة في مجال التردد تغطي كامل الفترة الأساسية من $-\pi$ إلى $+\pi$ يتم تغيير حدود التكامل لتصبح محصورة بين $[-\pi, +\pi]$ ، ونستبدل $x(\theta)$ بقيمتها الثابتة (1) :

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} 1 \cdot e^{jn\theta} d\theta$$

يتم مكاملة الدالة الأسية $e^{jn\theta}$ بالنسبة للمتغير θ

نصل إلى الشكل القياسي للنتيجة وهي دالة النبضة $x[n] = \delta[n]$

$$\theta = \omega T \text{ التردد الزاوي الرقمي}$$

$$\theta = 2\pi \text{ فإن } T = 1 \text{ و } \omega = 2\pi f$$

لذلك يتم حصر المجال بين $[-\pi, \pi]$ لتمثيل دورة ترددية واحدة كاملة

النتيجة النهائية هي: الطيف الترددي الثابت تماماً (المستوي) يقابله في الزمن نبضة حادة $\delta[n]$

IDTFT

$$x(\theta) \rightarrow x[n]$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} x(\theta) \cdot e^{jn\theta} d\theta$$

Homework:

Properties of DTFT

- Linearity
- Time shift
- Frequency shift



The end ...